

Zero-Bias Towers and Spectral Peeling in the Fabius–Rvachev System

Random-index q -calculus, normalized Laguerre–Pólya derivatives,
Thue–Morse release values, and logarithmic Gaussianization

Frontier research report for Vladimir Reshetnikov’s ProveIt corpus

Repository-relative novelty audit through commit
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Abstract

Let

$$Y = \sum_{j \geq 1} 2^{-j} U_j, \quad U_j \stackrel{\text{iid}}{\sim} \text{Unif}[-1, 1],$$

so that Y has Rvachev density up, while $X = (Y + 1)/2$ has the Fabius distribution function. This report studies the iterated zero-bias tower $Y = Y^{(0)}, Y^{(1)}, Y^{(2)}, \dots$, defined by $\mathbb{E}[Wf(W)] = \text{Var}(W)\mathbb{E}[f'(W^z)]$ at each symmetric level. The direction was selected only after a recursive search and manifest-level audit of the full Analysis/FabiusFunction/docs LaTeX corpus. General zero-biasing is classical; the claims of novelty below are deliberately repository-relative and concern its exact synthesis with the Fabius–Rvachev structures.

The tower admits a closed density kernel, an exact moment formula, and the factorization

$$Y^{(m)} \stackrel{d}{=} R_m B_m,$$

where R_m is the $2m$ -power-biased radius $|Y|$ and B_m is the first coordinate of a uniform point of the sphere S^{2m} . If

$$H(z) = \widehat{\text{up}}(i\sqrt{z}/\pi),$$

then every normalized derivative $H^{(m)}(z)/H^{(m)}(0)$ is exactly the square-moment generating transform of $Y^{(m)}$. Consequently, the Laguerre–Pólya and Pólya-frequency results already established for H propagate to a probabilistically meaningful hierarchy.

For the geometric family $Y_q = (1 - q) \sum_{j \geq 0} q^j U_j$, one zero-bias step replaces a random digit whose index is geometric with ratio q^2 . At level m , the numbers of replacements at the individual digits have an exact occupancy law. The probability that all m hits occur at distinct digits is

$$\frac{(2m - 1)!! m! (1 - q)^{2m} q^{m(m-1)}}{\mu_{2m}(q) 3^m (q^2; q^2)_m},$$

which exposes a new q -Pochhammer layer inside the zero-bias dynamics.

The derivative hierarchy also peels the arithmetic Fourier divisor. At the lattice frequency $2\pi N$, the original characteristic function has order $\nu_2(N) + 1$. Every tower step removes exactly one inherited order, and at the first nonzero level the released value has a closed product formula whose sign is $(-1)^{\nu_2(N) + 2 + s_2((u-1)/2)}$, where u is the odd part of N . Thus the Thue–Morse sequence controls not merely signs between zeros, but the first value exposed after a multiple zero is differentiated away.

Finally, the tower Gaussianizes every compact symmetric law whose support reaches its endpoints: $\sqrt{2m} Y^{(m)} \Rightarrow N(0, 1)$. For the Rvachev law the convergence is logarithmically slow. The exact negative-Laplace decomposition of the Fabius law yields a phase-dependent mod-Gamma law for $2m(1 - R_m)$ and the sharp transport scale

$$W_1\left(\sqrt{2m} Y^{(m)}, N(0, 1)\right) \sim \sqrt{\frac{2}{\pi}} \frac{\log(2m)}{2m \log 2}.$$

The report ends with conjectures on spectral non-reappearance, Bessel–Airy zero transitions, occupancy limit shapes, phase-resolved all-order expansions, and a concrete Lean formalization program. A fully commented Python program reproduces the tables, exact rational checks, Monte Carlo occupancy sampler, and figures.

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1 Scope, corpus audit, and claim discipline

The repository subtree examined for this report is `Analysis/FabiusFunction/docs`, recursively. The indexed tree contains more than ninety LaTeX sources and generated LaTeX fragments, in addition to Python, Lean, CSV, PDF, and archival assets. It is not a single article but a research library: a primary exposition, a mathematical glossary, a Lean walkthrough, a q -special-function survey, endpoint and inverse asymptotics, Fourier-decay reports, dyadic-comb studies, integration and transform volumes, Legendre and Lagrange representation projects, spectral-arithmetic reports, Stein–Koopman work, and several consolidated frontier compilations.

A title search would therefore be an inadequate duplication check. The audit used three layers:

1. recursive enumeration of the indexed `.tex` files;
2. the documentation audit, paper-coverage inventory, category READMEs, and the frontier manifest, which identify consolidations and superseded drafts;
3. direct text searches in the relevant current manuscripts for the terms *zero-bias*, *iterated zero-bias*, *power-biased radius*, *spectral peeling*, and *mod-Gamma*.

No occurrence of those organizing notions was located in the inspected tree. The closest antecedents are the Stein-kernel chapter of the Stein–Koopman report and the incoming total-positivity report. The former supplies a natural probabilistic neighborhood; the latter establishes the Laguerre–Pólya class of the square-root transform and the exact multiplicity of its arithmetic zeros. Both are used here as imported input rather than repackaged as new conclusions.

Claim convention. “New” means absent from the inspected repository snapshot and derived here. It is not a claim of worldwide priority. The zero-bias transform, replacement-of-a-summand coupling, and general higher-order distributional transforms are classical [6, 7, 4]. What is new in the present report is the exact Fabius–Rvachev specialization, its q -occupancy calculus, the probability interpretation of the normalized derivatives of H , the arithmetic release formulas, and the phase-resolved endpoint asymptotics assembled from the repository’s imported theorems.

Table 1: Separation between imported results and the results developed here.

Theme	Imported from the corpus	Developed in this report
Probability law	Uniform random-series representations, affine fixed points, smooth density, moments and cumulants	Random-index zero-bias recursion, complete iterated tower, spherical factorization
Entire transform	$H \in \mathcal{LP}$ of type I, PF_∞ coefficients, arithmetic zero divisor	$H^{(m)}/H^{(m)}(0)$ as a probability transform; interlacing tower and inherited Turán inequalities
Fourier arithmetic	Multiplicity $\nu_2(N) + 1$ and Thue–Morse signs of the original product	One-order-per-step peeling, exact released value and sign, divisor count at each level
q -structure	Geometric uniform laws, q -products, q -binomial identities	q^2 -geometric selector, occupancy Gibbs law, collision-free $(q^2; q^2)_m^{-1}$ formula
Endpoint asymptotics	Exact negative-Laplace quadratic term, Gamma–zeta periodic correction, lower-Lambert inverse theory	Power-biased mod-Gamma law, tower moment corrections, sharp leading Wasserstein rate

2 Normalization and imported analytic toolkit

2.1 The centered and uncentered laws

Let $U_1, U_2, \dots$ be independent uniform variables on $[-1, 1]$ and set

$$Y = \sum_{j=1}^{\infty} 2^{-j} U_j. \quad (1)$$

The series converges absolutely almost surely, is symmetric, and has support $[-1, 1]$. Its density is Rvachev's even up-function up. The affine variable

$$X = \frac{Y+1}{2} = \sum_{j=1}^{\infty} 2^{-j} V_j, \quad V_j = \frac{U_j+1}{2} \sim \text{Unif}[0, 1], \quad (2)$$

has distribution function F , the bounded Fabius function. Reflection gives $X \stackrel{d}{=} 1 - X$, and on $0 \leq x \leq 1$ the two normalizations satisfy

$$\text{up}(x) = F(1 - x). \quad (3)$$

Write

$$\mu_n = \mathbb{E}|Y|^n \quad (n \geq 0), \quad \mu_{2n} = \mathbb{E}Y^{2n}. \quad (4)$$

In particular,

$$\mu_0 = 1, \quad \mu_2 = \frac{1}{9}, \quad \mu_4 = \frac{19}{675}, \quad \mu_6 = \frac{583}{59535}. \quad (5)$$

The characteristic and moment generating functions are

$$\phi(t) = \mathbb{E}e^{itY} = \prod_{j=1}^{\infty} \text{sinc}\left(\frac{t}{2^j}\right), \quad (6)$$

$$M(t) = \mathbb{E}e^{tY} = \prod_{j=1}^{\infty} \frac{\sinh(t/2^j)}{t/2^j}, \quad (7)$$

where $\text{sinc } z = \sin z/z$ with its removable value at the origin. The even cumulants are

$$\kappa_{2r} = \frac{2^{2r} B_{2r}}{2r(2^{2r} - 1)}, \quad \kappa_{2r+1} = 0, \quad (8)$$

so Bell-polynomial recurrences recover all rational moments.

We shall also use the fixed-support geometric family

$$Y_q = (1 - q) \sum_{j=0}^{\infty} q^j U_j, \quad 0 < q < 1. \quad (9)$$

Its support is again $[-1, 1]$, and

$$\sigma_q^2 = \text{Var}(Y_q) = \frac{1 - q}{3(1 + q)}. \quad (10)$$

The Rvachev law is $q = 1/2$ after the harmless reindexing in (1).

2.2 The square-root entire transform

Using the repository's Fourier convention $\widehat{\text{up}}(z) = \int \text{up}(x) e^{-2\pi i z x} dx$, define

$$H(z) = \widehat{\text{up}}\left(\frac{i\sqrt{z}}{\pi}\right) = \prod_{j=0}^{\infty} \frac{\sinh(2^{-j}\sqrt{z})}{2^{-j}\sqrt{z}} = \sum_{n=0}^{\infty} \frac{4^n \mu_{2n}}{(2n)!} z^n. \quad (11)$$

The incoming total-positivity report proves that H belongs to the Laguerre–Pólya class of type I and that its coefficient sequence is PF_{∞} . Its zeros are

$$-\pi^2 N^2, \quad N = 1, 2, \dots, \quad (12)$$

with exact multiplicity

$$r(N) = \nu_2(N) + 1. \quad (13)$$

Here ν_2 is the 2-adic valuation. Relations

$$M(t) = H(t^2/4), \quad \phi(t) = H(-t^2/4) \quad (14)$$

will turn derivatives of H into probability transforms.

2.3 The negative-Laplace phase

Let

$$L_X(s) = \mathbb{E}e^{-sX} = \prod_{j=1}^{\infty} \frac{1 - e^{-s/2^j}}{s/2^j}, \quad \Lambda(s) = \log L_X(s), \quad (15)$$

and put $L = \log 2$. A central formalized result in the repository is the exact decomposition

$$\Lambda(s) = -\frac{(\log s)^2}{2L} + \frac{\log s}{2} + \Psi(\log_2 s) + E(s), \quad s > 0, \quad (16)$$

where Ψ is smooth and one-periodic, while $E(s)$ and its needed derivatives are exponentially small. With

$$\chi_k = \frac{2\pi i k}{L}, \quad (17)$$

the Fourier coefficients of Ψ are

$$\widehat{\Psi}(0) = 0, \quad \widehat{\Psi}(k) = -\frac{\Gamma(-\chi_k)\zeta(1-\chi_k)}{L} \quad (k \neq 0). \quad (18)$$

In particular, every nonzero Fourier mode is nonzero. Differentiating (16) gives

$$\Lambda'(s) = \frac{-\log s/L + 1/2 + \Psi'(\log_2 s)/L}{s} + \mathcal{O}(s^{-2}). \quad (19)$$

The periodic ripple is numerically tiny, but it survives at the first phase-resolved order and will reappear in the radius of the zero-bias tower.

3 One zero-bias step and the geometric digit selector

3.1 Definition and elementary identities

Definition 3.1 (Zero-bias transform). Let W have mean zero and variance $\sigma^2 \in (0, \infty)$. A random variable W^z has the W -zero-biased distribution if

$$\mathbb{E}[W f(W)] = \sigma^2 \mathbb{E}[f'(W^z)] \quad (20)$$

for every smooth test function for which the expectations exist.

Existence and uniqueness are classical [6]. Choosing $f(x) = x^{k+1}/(k+1)$ yields

$$\mathbb{E}[(W^z)^k] = \frac{\mathbb{E}W^{k+2}}{(k+1)\sigma^2}, \quad k \geq 0. \quad (21)$$

If W is symmetric, then W^z is symmetric as well, so the transform may be iterated without recentering.

For $U \sim \text{Unif}[-1, 1]$, direct integration gives

$$f_{U^z}(u) = \frac{3}{4}(1 - u^2)\mathbf{1}_{[-1,1]}(u), \quad (22)$$

or, equivalently,

$$U^z \stackrel{d}{=} 2B - 1, \quad B \sim \text{Beta}(2, 2). \quad (23)$$

Its characteristic function is

$$\beta_1(t) = 3 \frac{\sin t - t \cos t}{t^3}. \quad (24)$$

3.2 A q^2 -geometric replacement index

The standard independent-sum zero-bias coupling chooses one summand with probability proportional to its variance and replaces only that summand by its zero-biased version. For (9), the variance weights collapse to an exact geometric law.

Theorem 3.2 (Random-index zero bias for the geometric law). *Let $c_j = (1 - q)q^j$, and let J be independent of all digits with*

$$\mathbb{P}(J = j) = (1 - q^2)q^{2j}, \quad j = 0, 1, 2, \dots \quad (25)$$

Let V_j be independent copies of U^z . Then

$$Y_q^z \stackrel{d}{=} Y_q - c_J U_J + c_J V_J. \quad (26)$$

Equivalently, if all variables on the right are independent copies, then

$$Y_q^z \stackrel{d}{=} \begin{cases} q(Y_q')^z + (1 - q)U, & \text{with probability } q^2, \\ qY_q' + (1 - q)U^z, & \text{with probability } 1 - q^2. \end{cases} \quad (27)$$

Proof. The j th summand has variance $c_j^2/3$. Since

$$\sum_{j \geq 0} \frac{c_j^2}{3} = \frac{(1 - q)^2}{3(1 - q^2)} = \frac{1 - q}{3(1 + q)} = \sigma_q^2,$$

its selection probability is $c_j^2/(3\sigma_q^2) = (1 - q^2)q^{2j}$. This proves (26). Splitting $Y_q = qY_q' + (1 - q)U$, the two variance shares are q^2 and $1 - q^2$, which gives (27). Repeatedly taking the first branch until the second branch occurs produces exactly the geometric index in (25). $\square$

Corollary 3.3 (Characteristic function). *With $\phi_q(t) = \prod_{j \geq 0} \text{sinc}(c_j t)$,*

$$\phi_q^z(t) = \sum_{j \geq 0} (1 - q^2)q^{2j} \beta_1(c_j t) \prod_{\ell \neq j} \text{sinc}(c_\ell t). \quad (28)$$

Away from the zeros of the individual factors, this can be factored as

$$\phi_q^z(t) = \phi_q(t) \sum_{j \geq 0} (1 - q^2)q^{2j} \frac{3(1 - c_j t \cot(c_j t))}{(c_j t)^2}. \quad (29)$$

The unfactored formula (28) is the correct entire continuation at those zeros.

For $q = 1/2$, let $p_0 = \text{up}$ and let p_1 be the density of Y^z . Since $\mu_2 = 1/9$,

$$p_1(x) = 9 \int_{|x|}^1 y \text{up}(y) dy, \quad p_1'(x) = -9x \text{up}(x) \quad (-1 < x < 1). \quad (30)$$

The exact center value follows without numerical integration. Indeed, $\text{Var}(X) = \mu_2/4 = 1/36$ and $\mathbb{E}X = 1/2$, hence $\mathbb{E}X^2 = 5/18$. The moment bridge proved later in [Theorem 9.1](#) gives $\mathbb{E}|Y| = 5/18$, and therefore

$$p_1(0) = \frac{\mathbb{E}|Y|}{2\mu_2} = \frac{5}{4}. \quad (31)$$

4 The complete zero-bias tower

Definition 4.1 (Symmetric zero-bias tower). Let W be nondegenerate, symmetric, and compactly supported. Put $W^{(0)} = W$ and recursively

$$W^{(m+1)} = (W^{(m)})^z. \quad (32)$$

For the Rvachev law we write $Y^{(m)}$, and we denote its density by p_m .

The following theorem is independent of the special random series and is the main structural reason that the tower is tractable.

Theorem 4.2 (Moments of every tower level). *Let W be symmetric and let $m_r = \mathbb{E}|W|^r$. Then, for $m, r \geq 0$,*

$$\mathbb{E}[(W^{(m)})^{2r}] = \frac{(2m-1)!!(2r-1)!!}{(2m+2r-1)!!} \frac{m_{2m+2r}}{m_{2m}}, \quad (33)$$

with $(-1)!! = 1$. In particular,

$$\text{Var}(W^{(m)}) = \frac{m_{2m+2}}{(2m+1)m_{2m}}. \quad (34)$$

Proof. Formula (21), applied at level m , gives

$$\mathbb{E}[(W^{(m+1)})^{2r}] = \frac{\mathbb{E}[(W^{(m)})^{2r+2}]}{(2r+1) \text{Var}(W^{(m)})}.$$

Starting with $m = 0$, induction makes all intermediate moment ratios telescope. The remaining odd-number products are precisely the double factorials in (33). $\square$

4.1 Power-biased radius times a spherical coordinate

For $m \geq 1$, define the $2m$ -power-biased radius R_m by

$$\mathbb{E}g(R_m) = \frac{\mathbb{E}[|W|^{2m}g(|W|)]}{m_{2m}}. \quad (35)$$

Let B_m be independent of R_m and have density

$$b_m(u) = c_m(1-u^2)^{m-1} \mathbf{1}_{[-1,1]}(u), \quad c_m = \frac{\Gamma(m+1/2)}{\sqrt{\pi} \Gamma(m)}. \quad (36)$$

Equivalently, $(B_m + 1)/2 \sim \text{Beta}(m, m)$, and B_m is the first coordinate of a uniform point on the sphere $S^{2m} \subset \mathbb{R}^{2m+1}$.

Theorem 4.3 (Spherical factorization). *For every compactly supported symmetric W and every $m \geq 1$,*

$$W^{(m)} \stackrel{d}{=} R_m B_m. \quad (37)$$

Proof. The beta integral gives

$$\mathbb{E} B_m^{2r} = \frac{(2m-1)!!(2r-1)!!}{(2m+2r-1)!!}. \quad (38)$$

Independence and (35) therefore make the even moments of $R_m B_m$ equal to (33); all odd moments vanish. Both laws are compactly supported, hence moment-determinate. $\square$

Remark 4.4. The factorization cleanly separates two mechanisms. The beta coordinate is a universal spherical smoothing kernel, while the entire dependence on the original law is compressed into the endpoint-biased radius R_m . For the Rvachev law, the first mechanism produces Gaussian shape and the second is responsible for the anomalously slow logarithmic rate.

If W has an even density p on $[-1, 1]$, the factorization gives an explicit integral kernel.

Corollary 4.5 (Density kernel and differential descent). *For $m \geq 1$,*

$$p_m(x) = \frac{C_m}{m_{2m}} \int_{|x|}^1 y(y^2 - x^2)^{m-1} p(y) dy, \quad C_m = \frac{(2m-1)!!}{(2m-2)!!}. \quad (39)$$

For $0 < x < 1$,

$$p'_m(x) = -(2m-1) \frac{m_{2m-2}}{m_{2m}} x p_{m-1}(x). \quad (40)$$

Consequently, with $\mathcal{R} = x^{-1} \frac{d}{dx}$ interpreted on even smooth functions by continuity at zero,

$$\mathcal{R}^m p_m = (-1)^m \frac{(2m-1)!!}{m_{2m}} p. \quad (41)$$

Proof. The density of a product $R_m B_m$ is obtained by conditioning on $R_m = y$. The power-bias density contributes $2y^{2m}p(y)/m_{2m}$, the scale change contributes $1/y$, and (36) contributes $c_m(1 - x^2/y^2)^{m-1}$. Simplification gives $C_m = 2c_m$ and (39). Differentiation under the integral, including the $m = 1$ boundary case, gives (40). Iterating it telescopes the moment ratios and yields (41). $\square$

For the Rvachev density, every p_m is even, positive on $(-1, 1)$, compactly supported, and C^∞ -flat at ± 1 . Its center value is

$$p_m(0) = \frac{C_m \mu_{2m-1}}{2\mu_{2m}}. \quad (42)$$

4.2 Bessel mixture

Let

$$\mathcal{J}_m(z) = \Gamma\left(m + \frac{1}{2}\right) \left(\frac{2}{z}\right)^{m-1/2} J_{m-1/2}(z), \quad \mathcal{J}_m(0) = 1. \quad (43)$$

This is the characteristic function of B_m . The factorization therefore has the transform form

$$\phi_m(t) = \mathbb{E} \mathcal{J}_m(t R_m) = \frac{1}{\mu_{2m}} \mathbb{E} [|Y|^{2m} \mathcal{J}_m(t|Y|)]. \quad (44)$$

This formula will motivate the Bessel–Airy spectral conjecture in [Section 11](#).

5 Normalized Laguerre–Pólya derivatives as probability laws

Let

$$\mathcal{D}_t = \frac{1}{t} \frac{d}{dt} \quad (45)$$

act on even entire functions, with the removable value at $t = 0$. The zero-bias identity implies a radial derivative calculus.

Theorem 5.1 (Radial derivative formulas). *For the Rvachev tower,*

$$M_m(t) := \mathbb{E}e^{tY^{(m)}} = \frac{(2m-1)!!}{\mu_{2m}} \mathcal{D}_t^m M(t), \quad (46)$$

$$\phi_m(t) := \mathbb{E}e^{itY^{(m)}} = (-1)^m \frac{(2m-1)!!}{\mu_{2m}} \mathcal{D}_t^m \phi(t). \quad (47)$$

Equivalently,

$$M_m(t) = \frac{H^{(m)}(t^2/4)}{H^{(m)}(0)}, \quad (48)$$

$$\phi_m(t) = \frac{H^{(m)}(-t^2/4)}{H^{(m)}(0)}, \quad (49)$$

where

$$H^{(m)}(0) = \frac{2^m \mu_{2m}}{(2m-1)!!}. \quad (50)$$

Proof. Taking $f(x) = e^{tx}/t$ in the defining zero-bias identity gives $M_{W^z}(t) = M'_W(t)/(\text{Var}(W)t)$. At level m , use (34); the variances telescope and produce (46). The characteristic-function formula acquires one minus sign at each step. Finally, $\mathcal{D}_t H(\pm t^2/4) = \pm \frac{1}{2} H'(\pm t^2/4)$, so (14) turns the radial derivatives into ordinary derivatives of H . Evaluation at $t = 0$ gives (50). $\square$

Corollary 5.2 (A derivative hierarchy inside $\mathcal{LP}$). *Define*

$$H_m(z) = \frac{H^{(m)}(z)}{H^{(m)}(0)}. \quad (51)$$

Then

$$H_m(z) = \sum_{r=0}^{\infty} \frac{4^r \mathbb{E}[(Y^{(m)})^{2r}]}{(2r)!} z^r. \quad (52)$$

Every H_m is a type-I Laguerre–Pólya function with positive coefficients. Consequently:

1. every zero of ϕ_m is real;
2. the distinct negative zeros of consecutive H_m weakly interlace, becoming strictly interlacing after common multiplicities are removed;
3. the sequence in (52) is PF_{∞} for every m .

Proof. Ordinary derivatives preserve the Laguerre–Pólya class, and positivity of the coefficients is immediate from the moment interpretation. The real-zero and interlacing assertions follow from the standard approximation of Laguerre–Pólya functions by real-rooted polynomials. The type-I product criterion gives PF_{∞} coefficients. $\square$

The last statement extends the moment inequalities from the base law to the whole zero-bias hierarchy. For example, coefficient log-concavity and ordinary moment log-convexity give a two-sided inequality.

Corollary 5.3 (Two-sided tower moment inequality). *For $m \geq 0$ and $r \geq 1$, writing $m_{m,2r} = \mathbb{E}[(Y^{(m)})^{2r}]$,*

$$\frac{(2r)(2r-1)}{(2r+2)(2r+1)} m_{m,2r-2} m_{m,2r+2} \leq m_{m,2r}^2 \leq m_{m,2r-2} m_{m,2r+2}. \quad (53)$$

Both inequalities are strict for the Rvachev tower.

Remark 5.4. Equations (41) and (47) are a physical/Fourier dual pair. In physical space, m radial derivatives descend from p_m to the original density. In frequency space, m radial derivatives ascend from the original characteristic function to the tower. The same moment normalization appears on both sides.

6 Spectral peeling and Thue–Morse release values

The base characteristic function has the arithmetic zero lattice $t_N = 2\pi N$. Its multiplicity is $r(N) = \nu_2(N) + 1$. Differentiation removes these inherited multiplicities one at a time, but the first value after a zero is released contains more information than the multiplicity count alone.

For $N \geq 1$, write

$$r = \nu_2(N) + 1, \quad N = 2^{r-1}u, \quad u \text{ odd}, \quad (54)$$

and define the nonzero odd-tail product

$$A_u = \prod_{\ell=1}^{\infty} \operatorname{sinc}\left(\frac{\pi u}{2^\ell}\right) = \widehat{\operatorname{up}}(u/2). \quad (55)$$

The sign law already present in the repository gives

$$\operatorname{sgn}(A_u) = (-1)^{s_2((u-1)/2)}, \quad (56)$$

where $s_2(n)$ is the binary digit sum.

Theorem 6.1 (Exact local peeling). *Let $t_N = 2\pi N$, and let $0 \leq m < r$. Then t_N is a zero of ϕ_m of exact order $r - m$, and*

$$\phi_m(t) = (-1)^{m+1} \frac{(2m-1)!!}{\mu_{2m}} \frac{r!}{(r-m)!} \frac{A_u}{t_N^{r+m}} (t - t_N)^{r-m} + \mathcal{O}((t - t_N)^{r-m+1}). \quad (57)$$

At the release level $m = r$, the zero disappears and

$$\phi_r(2\pi N) = (-1)^{r+1} \frac{r!(2r-1)!!}{\mu_{2r}(2\pi N)^{2r}} A_u = (-1)^{r+1} \frac{(2r)!}{2^r \mu_{2r}(2\pi N)^{2r}} A_u. \quad (58)$$

Therefore

$$\operatorname{sgn} \phi_r(2\pi N) = (-1)^{r+1+s_2((u-1)/2)}. \quad (59)$$

Proof. Exactly the factors $j = 1, \dots, r$ in (6) vanish at t_N . For such j ,

$$\left. \frac{d}{dt} \operatorname{sinc}\left(\frac{t}{2^j}\right) \right|_{t=t_N} = \frac{(-1)^{N/2^{j-1}}}{2\pi N}. \quad (60)$$

Their signs multiply to

$$(-1)^{\sum_{j=1}^r N/2^{j-1}} = (-1)^{u(2^r-1)} = -1.$$

All remaining factors multiply to A_u . Hence

$$\phi(t) = -\frac{A_u}{t_N^r} (t - t_N)^r + \mathcal{O}((t - t_N)^{r+1}). \quad (61)$$

At a nonzero center t_N , the leading term of $\mathcal{D}_t^m$ is $t_N^{-m} \frac{d^m}{dt^m}$; all lower ordinary derivatives vanish to higher order. Substitution into (47) proves (57). Setting $m = r$ gives (58), and (56) gives the sign. $\square$

Table 2: First released values. The magnitude rapidly reflects both the high-order radial derivative and the distant frequency scale.

N	$r = \nu_2(N) + 1$	odd part u	sign	$\phi_r(2\pi N)$
1	1	1	+	$1.26244712565 \times 10^{-1}$
2	2	1	−	$-4.73360910162 \times 10^{-3}$
3	1	3	−	$-1.17031790882 \times 10^{-3}$
4	3	1	+	$2.01947080783 \times 10^{-5}$
5	1	5	−	$-7.89411694646 \times 10^{-5}$
6	2	3	+	$4.87573995722 \times 10^{-6}$
7	1	7	+	$4.86097382485 \times 10^{-6}$
8	4	1	−	$-8.38784456413 \times 10^{-9}$

6.1 The inherited divisor at level m

Define the total inherited positive-lattice multiplicity through frequency K by

$$D_m(K) = \sum_{N=1}^K \max\{\nu_2(N) + 1 - m, 0\}. \quad (62)$$

This deliberately counts only zeros forced by the original arithmetic divisor; it does not assume that a frequency cannot accidentally become a zero again at a later derivative level.

Proposition 6.2 (Closed inherited-divisor count). *Let $M = \lfloor K/2^m \rfloor$. Then*

$$D_m(K) = 2M - s_2(M). \quad (63)$$

Proof. A positive contribution requires $2^m \mid N$. Write $N = 2^m k$; the remaining multiplicity is $\nu_2(k) + 1$. Therefore

$$D_m(K) = \sum_{k=1}^M (\nu_2(k) + 1) = M + \nu_2(M!) = M + (M - s_2(M)),$$

using Legendre’s binary formula. □

6.2 Integer-base extension

The same mechanism is not uniquely dyadic. For an integer $b \geq 2$, set

$$Y_b = \left(1 - \frac{1}{b}\right) \sum_{j=0}^{\infty} b^{-j} U_j. \quad (64)$$

At

$$t_{b,N} = \frac{\pi b N}{b-1}, \quad (65)$$

the zero multiplicity is $r_b(N) = \nu_b(N) + 1$, where $\nu_b(N)$ is the largest r for which $b^r \mid N$. If $N = b^{r-1}u$ and

$$A_{b,u} = \prod_{\ell \geq 1} \text{sinc}(\pi u/b^\ell), \quad \varepsilon_{b,N} = (-1)^{u(b^r-1)/(b-1)}, \quad (66)$$

then the first released value is

$$\phi_{b,r}(t_{b,N}) = (-1)^r \varepsilon_{b,N} \frac{r!(2r-1)!!}{\mu_{2r}^{(b)} t_{b,N}^{2r}} A_{b,u}. \quad (67)$$

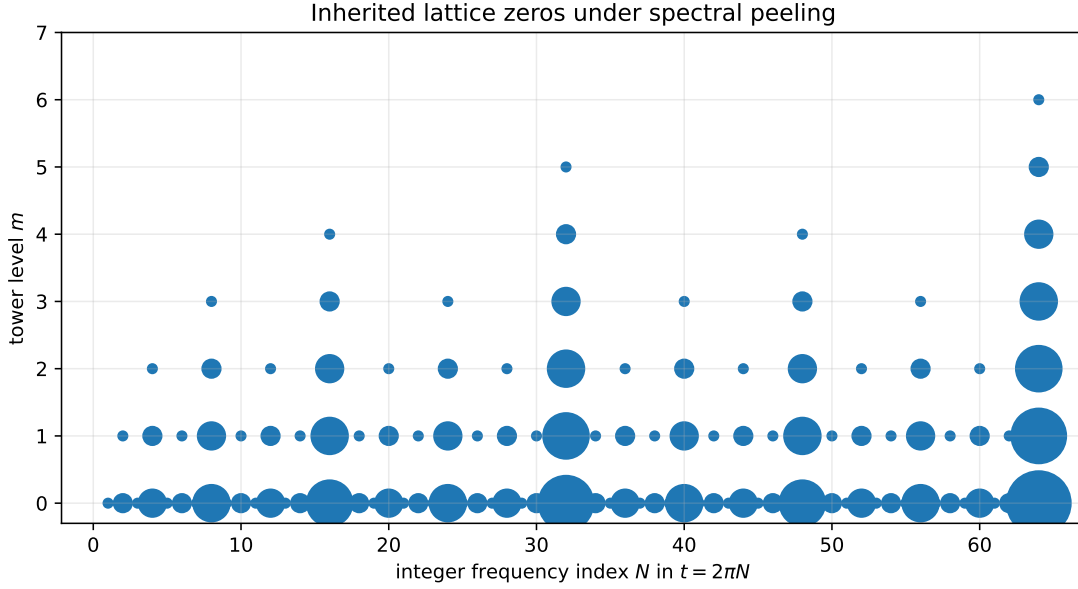


Figure 1: Inherited zero multiplicities for $1 \leq N \leq 64$. Marker area is proportional to the square of the remaining multiplicity. Each horizontal step retains only indices divisible by one additional power of two.

Moreover, with $M = \lfloor K/b^m \rfloor$ and $s_b(M)$ the base- b digit sum,

$$D_{b,m}(K) = \frac{bM - s_b(M)}{b-1}. \quad (68)$$

The proof repeats the dyadic argument; the identity $\sum_{j \geq 1} \lfloor M/b^j \rfloor = (M - s_b(M))/(b-1)$ replaces Legendre's binary formula.

7 Occupancy calculus and the q -Pochhammer layer

The one-step selector of [Theorem 3.2](#) is only the first row of a complete allocation law. At level m , a coordinate may have been zero-biased more than once. The resulting collision structure is exactly computable.

7.1 A general allocation theorem

Consider first a finite independent symmetric sum

$$W = X_1 + \cdots + X_d, \quad (69)$$

and write $\mu_{i,2k} = \mathbb{E}X_i^{2k}$ and $\mu_{W,2m} = \mathbb{E}W^{2m}$. Let $X_i^{(k)}$ denote the k th zero-bias level of X_i .

Theorem 7.1 (Zero-bias occupancy law). *For every $m \geq 0$, choose a weak composition $\mathbf{k} = (k_1, \dots, k_d)$ of m with probability*

$$\mathbb{P}(K_1 = k_1, \dots, K_d = k_d) = \frac{(2m-1)!! m!}{\mu_{W,2m}} \prod_{i=1}^d \frac{\mu_{i,2k_i}}{k_i! (2k_i-1)!!}. \quad (70)$$

Conditional on $\mathbf{K}$, take all transformed coordinates independently. Then

$$W^{(m)} \stackrel{d}{=} \sum_{i=1}^d X_i^{(K_i)}. \quad (71)$$

The same statement holds for an absolutely summable infinite series by truncation and weak convergence.

Proof. Let $M_i(t) = \mathbb{E}e^{tX_i}$ and use the derivation $\mathcal{D}_t = t^{-1} \frac{d}{dt}$. By [Theorem 5.1](#),

$$\mathcal{D}_t^k M_i(t) = \frac{\mu_{i,2k}}{(2k-1)!!} M_{i,k}(t), \quad (72)$$

where $M_{i,k}$ is the moment generating function of $X_i^{(k)}$. Apply the multinomial Leibniz rule to $\mathcal{D}_t^m \prod_i M_i(t)$, then multiply by $(2m-1)!!/\mu_{W,2m}$. The coefficients are exactly (70); evaluating at $t=0$ proves that they sum to one. The resulting product of component generating functions proves (71). $\square$

7.2 Geometric uniform digits

For $X_j = c_j U_j$, $c_j = (1-q)q^j$,

$$\mu_{j,2k} = \frac{c_j^{2k}}{2k+1}. \quad (73)$$

Thus (70) becomes

$$\mathbb{P}(\mathbf{K} = \mathbf{k}) = \frac{(2m-1)!! m!}{\mu_{2m}(q)} \prod_{j \geq 0} \frac{c_j^{2k_j}}{k_j!(2k_j+1)!!}, \quad \sum_j k_j = m. \quad (74)$$

Conditional on $K_j = k$, the j th digit is

$$U_j^{(k)} \stackrel{d}{=} 2B_{j,k} - 1, \quad B_{j,k} \sim \text{Beta}(k+1, k+1). \quad (75)$$

The local occupancy generating function is the elementary hyperbolic series

$$\sum_{k=0}^{\infty} \frac{x^k}{k!(2k+1)!!} = \frac{\sinh \sqrt{2x}}{\sqrt{2x}}. \quad (76)$$

Hence the partition function has the exact coefficient form

$$\frac{\mu_{2m}(q)}{(2m-1)!! m!} = [z^m] \prod_{j=0}^{\infty} \frac{\sinh(c_j \sqrt{2z})}{c_j \sqrt{2z}}. \quad (77)$$

This is the positive-coefficient counterpart of the alternating cumulant and sinc-product expansions.

Theorem 7.2 (Collision-free q -Pochhammer probability). *The probability that all m zero-bias hits fall on distinct digits is*

$$\mathbb{P}(K_j \in \{0, 1\} \forall j) = \frac{(2m-1)!! m!}{\mu_{2m}(q) 3^m} \frac{(1-q)^{2m} q^{m(m-1)}}{(q^2; q^2)_m}. \quad (78)$$

Proof. Restrict (74) to $k_j \in \{0, 1\}$. The remaining sum is 3^{-m} times the m th elementary symmetric function of $c_0^2, c_1^2, \dots$. Euler's identity gives

$$e_m(1, q^2, q^4, \dots) = \frac{q^{m(m-1)}}{(q^2; q^2)_m}. \quad (79)$$

Multiplication by $(1-q)^{2m}$ gives (78). $\square$

Table 3: Exact collision-free probabilities for the dyadic law $q = 1/2$. A single step is necessarily collision-free, but repeated hits rapidly concentrate on the leading digits.

m	exact probability	decimal
1	1	1
2	10/19	$5.2631578947 \times 10^{-1}$
3	70/583	$1.2006861063 \times 10^{-1}$
4	1400/132809	$1.0541454269 \times 10^{-2}$
5	15400/46840699	$3.2877391518 \times 10^{-4}$
6	14014000/4068990560161	$3.4440974470 \times 10^{-6}$
7	14014000/1204567303451311	$1.1634053124 \times 10^{-8}$

7.3 An exact sampler

Formula (74) gives an efficient sampler even at large tower levels. Truncate the digit set after J places, form suffix coefficient tables for the polynomials

$$W_j(z) = \sum_{k=0}^m \frac{c_j^{2k}}{k!(2k+1)!!} z^k, \quad (80)$$

and sample each K_j conditionally from the remaining coefficient. Then draw the beta digits in (75). For $q = 1/2$ and $J = 48$, the omitted spatial tail is below 4×10^{-15} . The accompanying Python file implements precisely this dynamic program; it does not use rare-event rejection from the $2m$ -power-biased radius.

8 Universal Gaussianization

The spherical factorization makes the limiting shape nearly universal. Normalize the endpoint of the support to one.

Theorem 8.1 (Universal compact-support Gaussianization). *Let W be symmetric, nondegenerate, supported in $[-1, 1]$, and suppose*

$$\mathbb{P}(|W| > 1 - \varepsilon) > 0 \quad \text{for every } \varepsilon > 0. \quad (81)$$

Then

$$\sqrt{2m} W^{(m)} \implies N(0, 1). \quad (82)$$

For every fixed $r \geq 0$ the moments converge as well:

$$\mathbb{E} \left[(\sqrt{2m} W^{(m)})^{2r} \right] = (2r - 1)!! \prod_{j=0}^{r-1} \frac{2m}{2m + 2j + 1} \frac{m_{2m+2r}}{m_{2m}} \longrightarrow (2r - 1)!! \quad (83)$$

Proof. Under the power-biased law (35), for every $\varepsilon > 0$,

$$\mathbb{P}(R_m \leq 1 - \varepsilon) = \frac{\mathbb{E}[|W|^{2m} \mathbf{1}_{\{|W| \leq 1 - \varepsilon\}}]}{m_{2m}} \longrightarrow 0.$$

Indeed, compare the numerator with a fixed positive amount of mass in $|W| > 1 - \varepsilon/2$. Thus $R_m \rightarrow 1$ in probability.

The spherical representation gives

$$B_m \stackrel{d}{=} \frac{G_1}{(G_1^2 + \dots + G_{2m+1}^2)^{1/2}}, \quad G_j \stackrel{\text{iid}}{\sim} N(0, 1).$$

Hence $\sqrt{2m} B_m \Rightarrow N(0, 1)$ by the law of large numbers. Slutsky's theorem and (37) prove (82). Formula (83) is (33) after scaling. The tilted radius concentrates at one, so $m_{2m+2r}/m_{2m} \rightarrow 1$. $\square$

Corollary 8.2 (Escape of characteristic zeros). *Let τ_m be the smallest positive zero of the Rvachev tower characteristic function ϕ_m . Then*

$$\frac{\tau_m}{\sqrt{2m}} \longrightarrow \infty. \quad (84)$$

Proof. The characteristic functions of $\sqrt{2m}Y^{(m)}$ converge uniformly on compact sets to $e^{-t^2/2}$, which has no real zero. Corollary 5.2 ensures that τ_m is a real zero, so it must leave every compact scaled interval. $\square$

The theorem gives convergence but no rate. For ordinary endpoint densities, $1 - R_m$ is typically of order $1/m$. The Rvachev density is infinitely flat at the endpoint, and this changes the scale to $(\log m)/m$. The next section makes that statement exact.

9 Fabius endpoint bias and phase-modulated mod-Gamma asymptotics

9.1 A bridge from absolute moments to the Fabius law

The endpoint radius is governed by the same negative-Laplace product that controls the small-argument Fabius asymptotics.

Lemma 9.1 (Moment–CDF bridge). *For every real $a > -1$,*

$$\mu_a = \mathbb{E}|Y|^a = \frac{2}{a+1} \mathbb{E}X^{a+1}. \quad (85)$$

Proof. Using (3), reflection of X , and Stieltjes integration by parts,

$$\begin{aligned} \mu_a &= 2 \int_0^1 r^a F(1-r) \, dr = 2 \int_0^1 (1-x)^a F(x) \, dx \\ &= \frac{2}{a+1} \mathbb{E}(1-X)^{a+1} = \frac{2}{a+1} \mathbb{E}X^{a+1}. \end{aligned}$$

$\square$

For $n > 0$, let $\mathcal{R}_n$ denote the n -power-biased radius:

$$\mathbb{E}g(\mathcal{R}_n) = \frac{\mathbb{E}[|Y|^n g(|Y|)]}{\mu_n}. \quad (86)$$

Thus the radius in the level- m factorization is $R_m = \mathcal{R}_{2m}$. Put

$$T_n = n(1 - \mathcal{R}_n). \quad (87)$$

After the change of variable $x = 1 - r$, the Laplace transform of T_n is

$$\mathbb{E}e^{-uT_n} = \frac{I_n(u)}{I_n(0)}, \quad I_n(u) = \int_0^1 e^{-ux} (1-x)^n F(x) \, dx. \quad (88)$$

The auxiliary Laplace integral

$$J(s) = \int_0^1 e^{-sx} F(x) \, dx = \frac{L_X(s) - e^{-s}}{s} \quad (89)$$

follows by integration by parts.

Lemma 9.2 (Power tilt versus exponential tilt). *Uniformly for u in a fixed compact subset of $(-1, \infty)$,*

$$\frac{I_n(u)}{I_n(0)} = \frac{J((1+u)n)}{J(n)} \left(1 + \mathcal{O}\left(\frac{(\log n)^2}{n}\right) \right). \quad (90)$$

The same estimate remains valid after any fixed number of derivatives with respect to u in a neighborhood of the origin.

Proof architecture. The inequality $\log(1-x) \leq -x$ gives $(1-x)^n \leq e^{-nx}$. On $0 \leq x \leq 1/2$,

$$0 \leq e^{-nx} - (1-x)^n \leq Cnx^2e^{-nx}.$$

For the measure proportional to $e^{-sx}F(x)dx$, derivatives of (16) and (89) give

$$\mathbb{E}_s x^k = \mathcal{O}_k((\log s/s)^k).$$

Consequently, the relative error made by replacing $(1-x)^n$ with e^{-nx} is $O((\log n)^2/n)$. The region $x > 1/2$ is exponentially small relative to $J(n)$, whose logarithm has only quadratic-logarithmic decay. Inserting fixed powers of nx proves the differentiated estimates. $\square$

9.2 A periodic mod-Gamma law

Define

$$\alpha_n = \log_2 n + \frac{1}{2}, \quad \theta_n = \log_2 n, \quad (91)$$

and, for $u > -1$,

$$\mathcal{R}_\theta(u) = \exp\left[-\frac{\log^2(1+u)}{2L} + \Psi(\theta + \log_2(1+u)) - \Psi(\theta)\right]. \quad (92)$$

The notation $\mathcal{R}_\theta$ here denotes the residue function, not the power-biased radius $\mathcal{R}_n$; the distinction is always clear from the argument.

Theorem 9.3 (Phase-modulated mod-Gamma endpoint law). *Uniformly for u in a fixed compact subset of $(-1, \infty)$,*

$$\boxed{\mathbb{E}e^{-uT_n} = (1+u)^{-\alpha_n} \mathcal{R}_{\theta_n}(u) \left(1 + \mathcal{O}\left(\frac{(\log n)^2}{n}\right) \right)}. \quad (93)$$

In particular, along every sequence for which the fractional part of $\log_2 n$ tends to θ ,

$$(1+u)^{\alpha_n} \mathbb{E}e^{-uT_n} \longrightarrow \mathcal{R}_\theta(u). \quad (94)$$

Proof. By Theorem 9.2 and (89), the logarithm of the left side is

$$\Lambda((1+u)n) - \Lambda(n) - \log(1+u) + \mathcal{O}((\log n)^2/n).$$

Put $a = \log(1+u)$. Substituting (16) gives

$$\begin{aligned} & -\frac{(\log n + a)^2 - (\log n)^2}{2L} + \frac{a}{2} - a + \Psi(\theta_n + a/L) - \Psi(\theta_n) \\ & = -\left(\frac{\log n}{L} + \frac{1}{2}\right)a - \frac{a^2}{2L} + \Psi(\theta_n + \log_2(1+u)) - \Psi(\theta_n). \end{aligned}$$

Exponentiation proves (93). $\square$

The dominant factor $(1 + u)^{-\alpha_n}$ is the Laplace transform of a $\text{Gamma}(\alpha_n, 1)$ variable. The Gamma–zeta periodic function survives as a bounded phase cocycle rather than disappearing in the limit.

Corollary 9.4 (Endpoint mean, variance, and central limit theorem). *As $n \rightarrow \infty$,*

$$\mathbb{E}T_n = \alpha_n - \frac{\Psi'(\theta_n)}{L} + \mathcal{O}\left(\frac{(\log n)^2}{n}\right), \quad (95)$$

$$\text{Var}(T_n) = \alpha_n - \frac{1}{L} - \frac{\Psi'(\theta_n)}{L} + \frac{\Psi''(\theta_n)}{L^2} + \mathcal{O}\left(\frac{(\log n)^2}{n}\right). \quad (96)$$

Moreover,

$$\frac{T_n - (\alpha_n - \Psi'(\theta_n)/L)}{\sqrt{\alpha_n}} \Rightarrow N(0, 1). \quad (97)$$

Proof. Differentiate the logarithm of (93) at $u = 0$. The first two derivatives of $a = \log(1 + u)$ are 1 and -1 , which gives (95) and (96). For (97), evaluate the moment generating function at an argument of order $\alpha_n^{-1/2}$ and expand $\log(1 + u)$. The residue contributes only bounded cumulants, while the Gamma factor contributes α_n to every fixed cumulant before standardization. $\square$

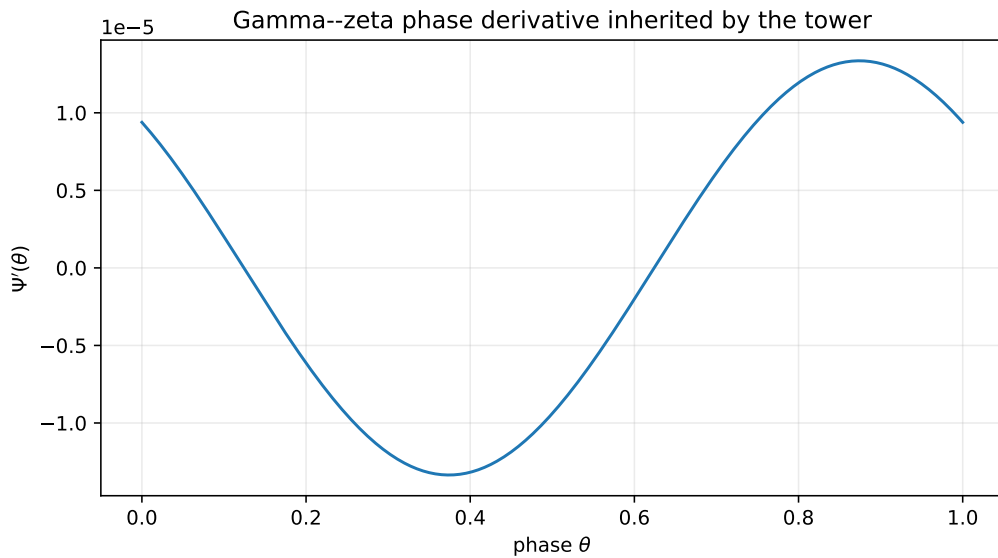


Figure 2: The periodic derivative Ψ' computed from (18). Its amplitude is only about 1.4×10^{-5} , but it enters the exact first phase-resolved term of the endpoint mean and every fixed moment ratio.

9.3 High-moment and tower-moment consequences

Since $\mathbb{E}\mathcal{R}_n^k = \mu_{n+k}/\mu_n$, the endpoint law immediately gives fixed-shift moment ratios.

Corollary 9.5 (Phase-resolved high-moment ratios). *For every fixed positive integer k ,*

$$1 - \frac{\mu_{n+k}}{\mu_n} = \frac{k}{n} \left(\log_2 n + \frac{1}{2} - \frac{\Psi'(\log_2 n)}{L} \right) + \mathcal{O}_k\left(\frac{(\log n)^2}{n^2}\right). \quad (98)$$

In particular,

$$1 - \frac{\mu_{2m+2}}{\mu_{2m}} \sim \frac{\log(2m)}{m \log 2}. \quad (99)$$

Proof. Write $\mathcal{R}_n = 1 - T_n/n$ and expand $1 - (1 - T_n/n)^k$. By (95), the linear term gives the displayed coefficient. The second moment is $O((\log n)^2)$ by (96), so all remaining terms contribute $O_k((\log n)^2/n^2)$. $\square$

Combining (98) with (83) gives a phase-resolved rate for every fixed tower moment.

Corollary 9.6 (First correction to Gaussian tower moments). *Put $n = 2m$. For fixed $r \geq 1$,*

$$\begin{aligned} & \frac{\mathbb{E}[(\sqrt{2m}Y^{(m)})^{2r}]}{(2r-1)!!} \\ &= 1 - \frac{2r \log_2 n + r + r^2 - 2r\Psi'(\log_2 n)/L}{n} + \mathcal{O}_r\left(\frac{(\log n)^2}{n^2}\right). \end{aligned} \quad (100)$$

For the variance this specializes to

$$1 - \text{Var}(\sqrt{2m}Y^{(m)}) = \frac{2 \log_2(2m) + 2 - 2\Psi'(\log_2(2m))/L}{2m} + \mathcal{O}\left(\frac{(\log m)^2}{m^2}\right). \quad (101)$$

Proof. For fixed r , (98) gives

$$\frac{\mu_{n+2r}}{\mu_n} = 1 - \frac{2r}{n} \left(\log_2 n + \frac{1}{2} - \frac{\Psi'(\log_2 n)}{L} \right) + \mathcal{O}_r((\log n)^2/n^2).$$

The universal beta factor satisfies

$$\prod_{j=0}^{r-1} \frac{n}{n+2j+1} = 1 - \frac{r^2}{n} + \mathcal{O}_r(n^{-2}).$$

Multiplying proves the claim. $\square$

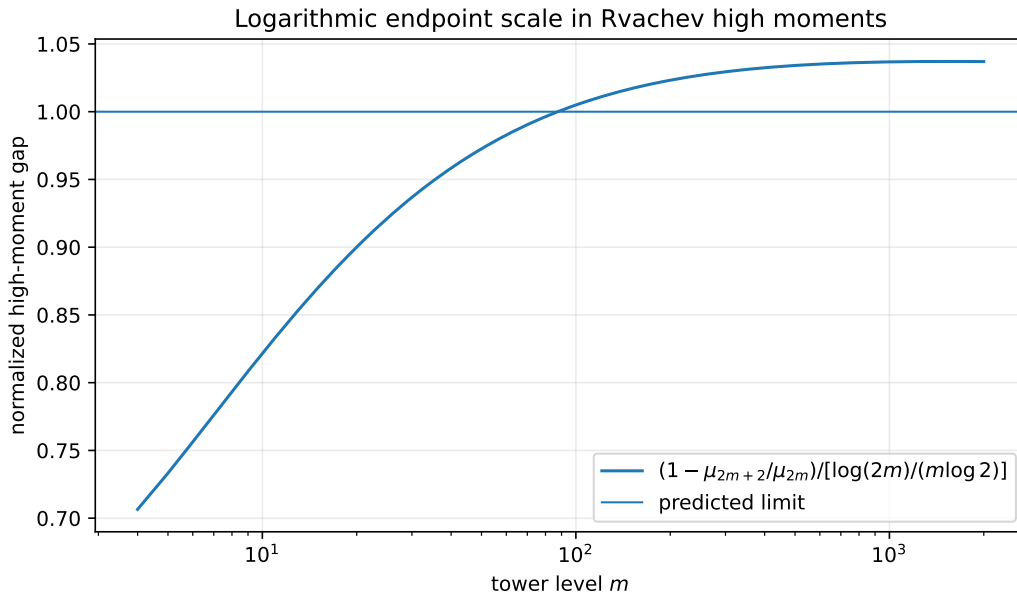


Figure 3: The normalized gap in (99), computed by the positive moment recurrence in Section A. The slow drift above one is consistent with the explicit constant term in (98); the periodic term is much smaller on this scale.

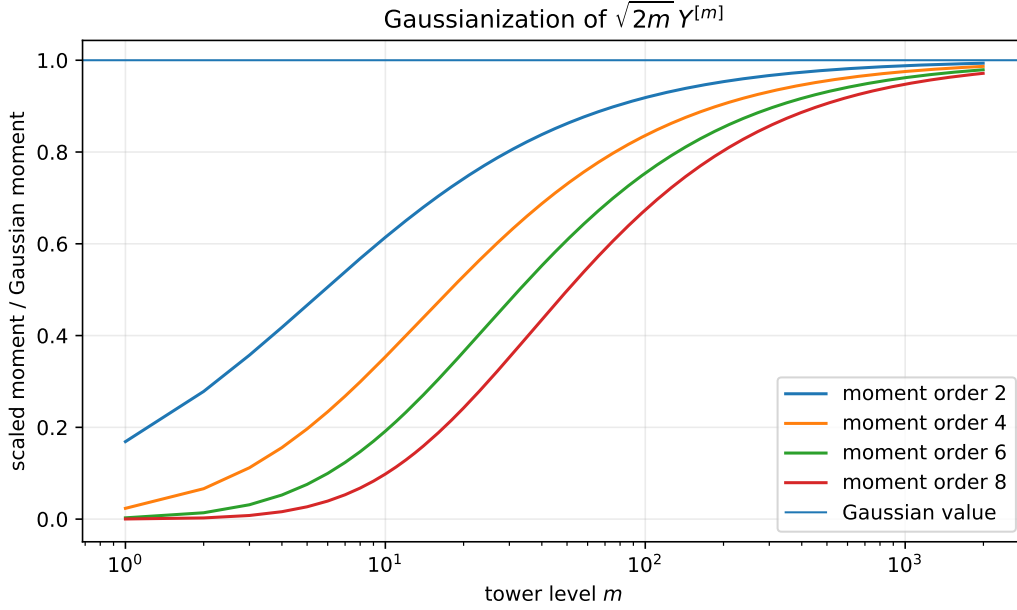


Figure 4: The first four even moments of $\sqrt{2m}Y^{[m]}$, divided by their Gaussian values. Higher moments converge more slowly, as predicted by the factor $2r \log_2(2m)/(2m)$ in (100).

9.4 A sharp leading Wasserstein rate

Let $Z \sim N(0, 1)$ and set

$$Z_m = \sqrt{2m}Y^{[m]}, \quad G_m = \sqrt{2m}B_m. \quad (102)$$

The factorization couples $Z_m = R_m G_m$ with R_m independent of G_m .

Lemma 9.7 (Spherical-coordinate normal approximation). *For the beta coordinate in (36),*

$$W_1(G_m, Z) = \mathcal{O}(m^{-1}). \quad (103)$$

Moreover,

$$\mathbb{E}|G_m| = \sqrt{2m} \frac{c_m}{m} = \sqrt{\frac{2}{\pi}} \left(1 - \frac{1}{8m} + O(m^{-2}) \right). \quad (104)$$

Proof architecture. The density of G_m is

$$g_m(x) = \frac{c_m}{\sqrt{2m}} \left(1 - \frac{x^2}{2m} \right)_+^{m-1}.$$

Stirling's expansion for c_m and a Taylor expansion of the logarithm show that the first absolute moment of the signed density difference from the standard normal is $O(1/m)$; this bounds W_1 . The exact integral $\mathbb{E}|B_m| = c_m/m$ gives (104). $\square$

Theorem 9.8 (Sharp leading transport scale). *For the Rvachev zero-bias tower,*

$$W_1(Z_m, Z) \sim \sqrt{\frac{2}{\pi} \frac{\log(2m)}{2m \log 2}}. \quad (105)$$

Proof. Let

$$d_m = \mathbb{E}(1 - R_m) = 1 - \frac{\mu_{2m+1}}{\mu_{2m}}.$$

By (98),

$$d_m = \frac{\log_2(2m) + 1/2 - \Psi'(\log_2(2m))/L}{2m} + O((\log m)^2/m^2). \quad (106)$$

The factorization coupling gives

$$W_1(Z_m, G_m) \leq \mathbb{E}|G_m|(1 - R_m) = \mathbb{E}|G_m|d_m.$$

Together with Theorem 9.7, this yields

$$W_1(Z_m, Z) \leq \sqrt{\frac{2}{\pi}}d_m + O(m^{-1}). \quad (107)$$

For the reverse bound, the function $x \mapsto |x|$ is 1-Lipschitz, so

$$\begin{aligned} W_1(Z_m, Z) &\geq \mathbb{E}|Z| - \mathbb{E}|Z_m| \\ &= \sqrt{\frac{2}{\pi}} - \mathbb{E}|G_m|\mathbb{E}R_m = \sqrt{\frac{2}{\pi}}d_m + O(m^{-1}). \end{aligned}$$

Since $d_m \sim \log(2m)/(2m \log 2)$ dominates m^{-1} , the upper and lower bounds have the same leading term. $\square$

Remark 9.9 (Where Lambert W went). The inverse Fabius endpoint problem uses a lower-Lambert variable because one must invert the relation between a tiny probability and its saddle scale. In the high-moment problem the saddle parameter is already supplied externally: it is the moment order n . The same quadratic logarithm and the same Gamma–zeta phase therefore appear directly at $\log_2 n$, without an additional Lambert inversion. In this precise sense the tower is the *deparametrized Laplace-side companion* of the lower- W_{-1} endpoint asymptotics.

10 Numerical experiments

The accompanying program `zero_bias_tower_experiments.py` performs both exact and floating-point checks. It uses no external data. Its principal computations are:

1. exact rational moments from Bernoulli cumulants and the complete Bell recurrence;
2. a positive, cancellation-free moment recurrence derived from $Y \stackrel{d}{=} (Y + U)/2$;
3. exact collision probabilities and release signs;
4. evaluation of Ψ' from the exponentially convergent Gamma–zeta Fourier series;
5. occupancy-based Monte Carlo sampling of tower densities;
6. deterministic moment and spectral-peeling plots.

10.1 Selected high-moment data

The normalized gap need not approach one monotonically. The $1/n$ constant term is still visible when divided by the leading $(\log n)/n$ term, and the phase correction is roughly five orders of magnitude smaller than unity.

10.2 Occupancy sampling

The variance check written by the script compares Monte Carlo estimates with (34). Because the displayed normalization is $\sqrt{2m}$ rather than exact unit variance, the low-level curves are visibly narrower than the normal. This is predicted quantitatively by (101), not a sampling defect.

Table 4: Even-moment ratios and the variance of the scaled tower.

m	μ_{2m+2}/μ_{2m}	normalized gap in (99)	$\text{Var}(\sqrt{2m}Y^{(m)})$
10	0.644909548417	0.821601942	0.614199569921
20	0.760481812449	0.900118090	0.741933475560
50	0.870759209336	0.972633866	0.862137831026
100	0.923183207674	1.004948163	0.918590256392
200	0.955776340501	1.023239131	0.953392858355
500	0.979388268891	1.034124888	0.978409859032
1000	0.988631011017	1.036769344	0.988136942545
1500	0.992013838782	1.037096739	0.991683277689

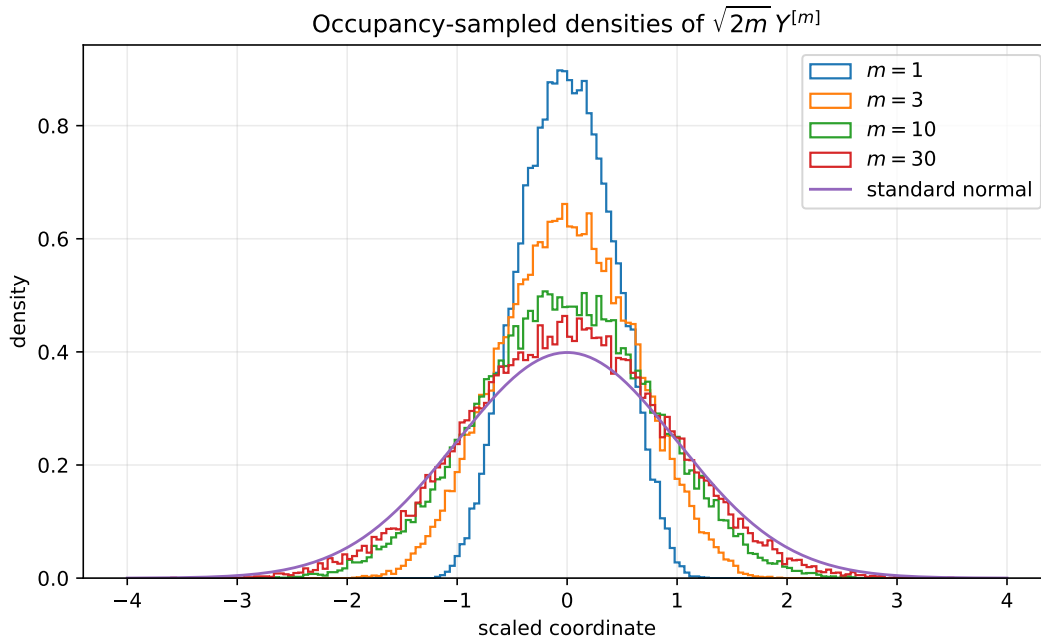


Figure 5: Occupancy-sampled densities of $\sqrt{2m}Y^{(m)}$ compared with the standard normal density. The plots use the exact conditional beta digits from (75); no endpoint rejection sampling is involved.

11 Conjectures and frontier problems

The exact results above expose several problems that are both structurally natural and sufficiently constrained to support serious attack.

Conjecture 11.1 (No spectral reappearance). *Let $N \geq 1$ and $r = \nu_2(N) + 1$. Then*

$$\phi_m(2\pi N) \neq 0 \quad \text{for every } m \geq r. \quad (108)$$

Equivalently, once the inherited lattice zero has been released at level r , it never reappears as an accidental zero of a later normalized derivative of H .

The statement is stronger than ordinary interlacing. Writing $H(z) = (1 + z/(\pi^2 N^2))^r Q_N(z)$ reduces it to the nonvanishing of $Q_N^{(m-r)}(-\pi^2 N^2)$. Total positivity may impose a sign regularity on these derivatives that is invisible in the unnormalized product.

Conjecture 11.2 (Bessel–Airy first-zero transition). *Let $\tau_{m,k}$ be the k th positive zero of ϕ_m , and let $j_{m-1/2,k}$ be the k th positive zero of $J_{m-1/2}$. For fixed k ,*

$$\tau_{m,k} = j_{m-1/2,k} + \frac{1}{2} \left(\log_2(2m) + \frac{1}{2} - \frac{\Psi'(\log_2(2m))}{L} \right) + o(1). \quad (109)$$

The first term is forced by the spherical beta factor. At the Bessel turning point, fluctuations of R_m occur on a scale much smaller than the $m^{1/3}$ Airy scale, while the mean contraction shifts the argument by approximately $j_{m-1/2,k} \mathbb{E}(1 - R_m)$. A proof should combine uniform turning point asymptotics for Bessel functions with the mod-Gamma law of [Theorem 9.3](#).

Conjecture 11.3 (Geometric occupancy limit shape). *For fixed $q \in (0, 1)$ and $c_j = (1 - q)q^j$, let $K_j^{(m)}$ have the occupancy law (74). Then, for each fixed J ,*

$$\left(\frac{K_0^{(m)}}{m}, \dots, \frac{K_J^{(m)}}{m} \right) \longrightarrow (c_0, \dots, c_J) \quad (110)$$

in probability. After centering and scaling by $\sqrt{m}$, the finite vectors converge to a Gaussian law with covariance

$$\frac{1}{2} (c_i \delta_{ij} - c_i c_j)_{0 \leq i, j \leq J}. \quad (111)$$

A saddle parameter $z \sim 2m^2$ in the product (77) predicts $\mathbb{E}K_j \sim c_j m$ and grand-canonical variance $c_j m/2$. Conditioning on the total occupancy produces the rank-one covariance correction in (111). Multivariate analytic combinatorics should make this route rigorous.

Conjecture 11.4 (All-order phase-resolved Gaussianization). *For every fixed r , the ratio in (100) admits a complete expansion*

$$\frac{\mathbb{E}[(\sqrt{2m}Y^{(m)})^{2r}]}{(2r-1)!!} \sim 1 + \sum_{j \geq 1} \frac{P_{r,j}(\log m, \{\log_2 m\})}{m^j}, \quad (112)$$

where $P_{r,j}$ is polynomial in $\log m$ and smooth periodic in the dyadic phase. Its periodic coefficients are differential polynomials in Ψ . An analogous expansion holds for smooth test functions against the Gaussian law.

The exact decomposition (16), the power-to-Laplace replacement, and Bell-polynomial differentiation make this conjecture highly structured. The main work is uniform bookkeeping rather than discovery of new saddle points.

Conjecture 11.5 (Monotone standardized Stein discrepancy). *Let $\tilde{Y}_m = Y^{(m)}/\sqrt{\text{Var}(Y^{(m)})}$. A natural quadratic Stein discrepancy, and possibly the relative entropy to the standard normal, is eventually strictly decreasing in m .*

Zero bias fixes the normal distribution after variance normalization, and the tower is an explicit deterministic orbit toward that fixed point. The spherical factorization and the density descent (40) provide unusually concrete tools for testing contractivity.

11.1 Further directions

Integer and nonintegral geometric bases. The release formula (67) and divisor count (68) should be combined with the arbitrary-base negative-Laplace decomposition. For integer b , one expects a $\log_b n$ -periodic mod-Gamma residue and a base- b digit-sign release law. For general q , the occupancy theorem is already exact; Mellin analysis should supply the corresponding period $\log(1/q)$.

Ultraspherical and Legendre synthesis. The kernel $(1 - u^2)^{m-1}$ is the Gegenbauer weight with parameter $m - 1/2$. Thus the tower defines a triangular transform from the repository’s Legendre coefficients of up to ultraspherical coefficients of p_m . An explicit coefficient map could close a new representation loop and may be better conditioned than repeated differentiation of the Legendre series.

Dyadic exact values. Formula (39) expands into finitely many weighted primitives of up. At dyadic x , the repository’s exact primitive and moment machinery should therefore yield rational values of $p_m(x)$ and its derivatives. Determining denominator valuations as a function of (m, x) is a concrete arithmetic project.

Multivariate masks. For compactly supported refinable densities generated by vector-valued uniform series, directional zero bias should replace a variance-weighted coordinate and produce a spherical factor in the chosen direction. The analogue of spectral peeling would act on hyperplane zero arrangements of the Fourier mask.

12 Lean formalization plan

The new results divide naturally into small reusable modules. A practical formalization order is:

1. define zero bias for symmetric compactly supported probability measures through the test-function identity and prove the polynomial moment rule (21);
2. formalize the independent-sum replacement theorem and specialize it to the q^2 -geometric selector (25);
3. define the symmetric tower and prove the moment formula (33);
4. construct the beta coordinate B_m , prove its even moments, and identify the product law $R_m B_m$ by compact moment determinacy;
5. prove the radial derivative identities and connect them to the existing entire transform H ;
6. reuse the current zero-multiplicity and Thue–Morse sign theorems to prove (58) and (63);
7. formalize the finite occupancy theorem with formal power series, then pass to the geometric infinite series by the repository’s existing truncation infrastructure;
8. import the negative-Laplace decomposition to establish the leading endpoint mean, high-moment ratio, and Wasserstein asymptotic.

The first six items require little asymptotic infrastructure and are plausible near-term targets. The mod-Gamma theorem needs a small library for uniform Laplace-tilt moment bounds, but its analytic input is already present in the formal development.

13 Conclusion

The zero-bias transform reveals a coherent layer that was not explicit in the inspected Fabius–Rvachev corpus. It turns the derivatives of the Laguerre–Pólya transform into genuine probability laws; it converts the geometric random series into a q^2 -selected digit replacement process; it organizes repeated replacements by a positive occupancy partition function; and it differentiates the arithmetic Fourier divisor one order at a time. The first value released from a multiple zero remembers the odd tail of the sinc product and hence the Thue–Morse sign.

At large depth, a universal spherical coordinate forces Gaussian shape. The Fabius endpoint nevertheless leaves a precise fingerprint: a power-biased radius with Gamma-scale fluctuations of

size $\log_2 m$, a nonconstant Gamma–zeta phase residue, and a Wasserstein delay of order $(\log m)/m$. This connects the probability, q-series, entire-function, Fourier-arithmetic, and Lambert-asymptotic parts of the repository without duplicating their existing central theorems.

A Stable moment recurrence and exact checks

The cumulant recurrence from (8) is exact but alternating. For high degrees, the distributional fixed point

$$Y \stackrel{d}{=} \frac{Y' + U}{2} \quad (113)$$

provides a positive recurrence. Taking the $2r$ th moment and isolating the self-term gives

$$\mu_{2r} = \frac{1}{2^{2r} - 1} \sum_{j=0}^{r-1} \binom{2r}{2j} \frac{\mu_{2j}}{2r - 2j + 1}. \quad (114)$$

Every summand is nonnegative. Evaluating the normalized binomial weights via log-Gamma functions makes the recurrence stable in ordinary double precision through at least the several thousand moments needed by the figures.

For low degrees, the exact values generated by the Bell recurrence begin

$$\begin{aligned} \mu_0 &= 1, & \mu_2 &= \frac{1}{9}, & \mu_4 &= \frac{19}{675}, \\ \mu_6 &= \frac{583}{59535}, & \mu_8 &= \frac{132809}{32531625}, & \mu_{10} &= \frac{46840699}{24405225075}. \end{aligned}$$

These values independently normalize the released-value and collision tables.

B Occupancy dynamic program

For a fixed tower level m , truncate after J geometric digits and define

$$w_j(k) = \frac{c_j^{2k}}{k!(2k+1)!!}, \quad 0 \leq k \leq m. \quad (115)$$

Let $S(j, r)$ be the coefficient of z^r in $\prod_{\ell=j}^{J-1} \sum_{k=0}^m w_\ell(k) z^k$. Backward convolution computes all $S(j, r)$ in $O(Jm^2)$ operations. Given that r hits remain at digit j , choose k with probability

$$\mathbb{P}(K_j = k \mid r \text{ remain}) = \frac{w_j(k) S(j+1, r-k)}{S(j, r)}. \quad (116)$$

After the occupancy vector is sampled, draw the independent beta digits from (75). The recurrence

$$\frac{w_j(k)}{w_j(k-1)} = \frac{c_j^2}{k(2k+1)} \quad (117)$$

avoids factorial overflow.

C Files generated by the companion program

A default run writes:

- `data/exact_even_moments.csv`;

- `data/high_moment_ratios.csv`;
- `data/collision_free_probabilities.csv`;
- `data/released_fourier_values.csv`;
- `data/tower_moment_convergence.csv`;
- `data/occupancy_sampler_checks.csv`;
- the PDF and PNG versions of all figures used in this report.

The source is extensively commented and accepts command-line options for the maximum moment degree, Monte Carlo sample count, and random seed.

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